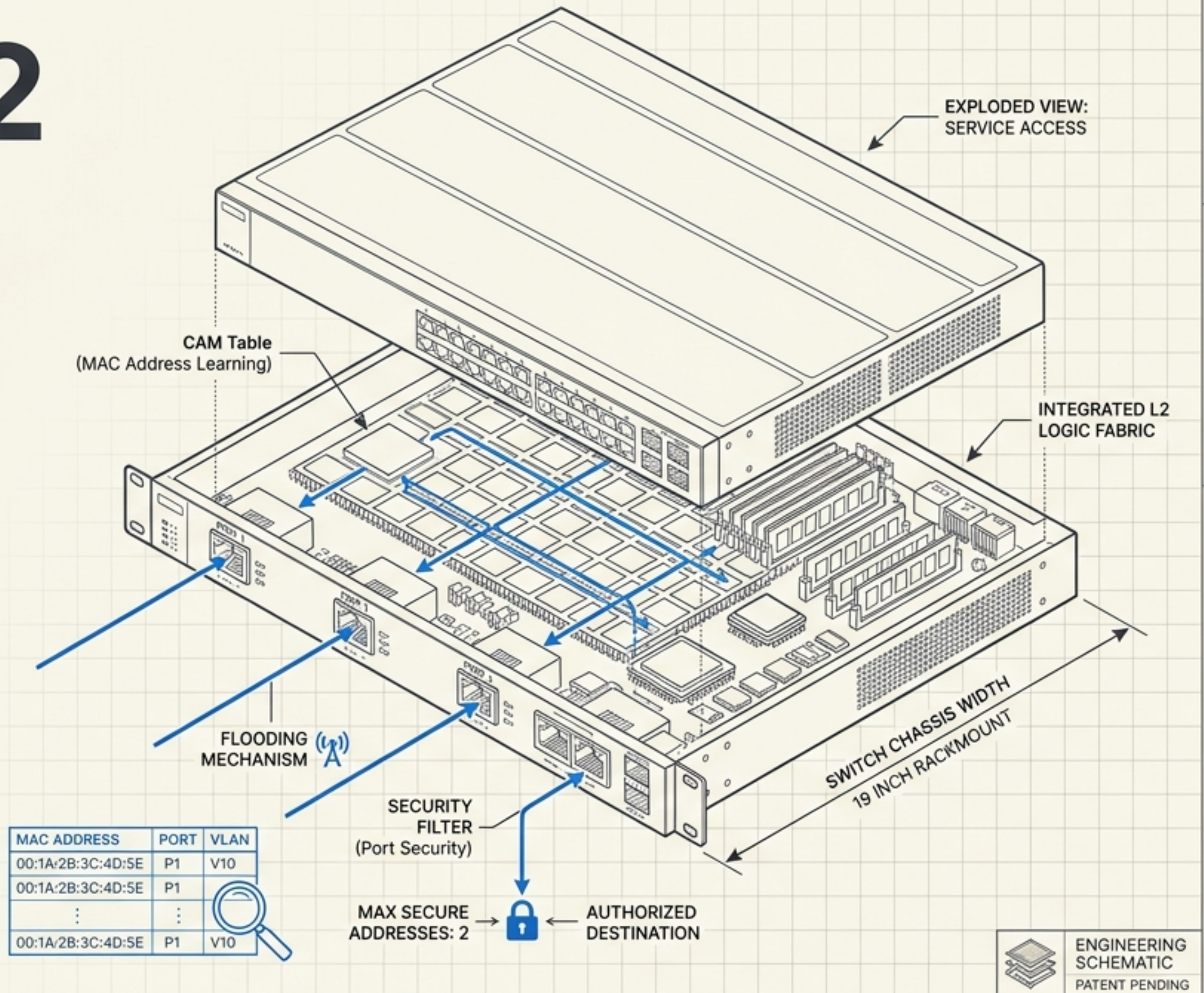


The Layer 2 Engine

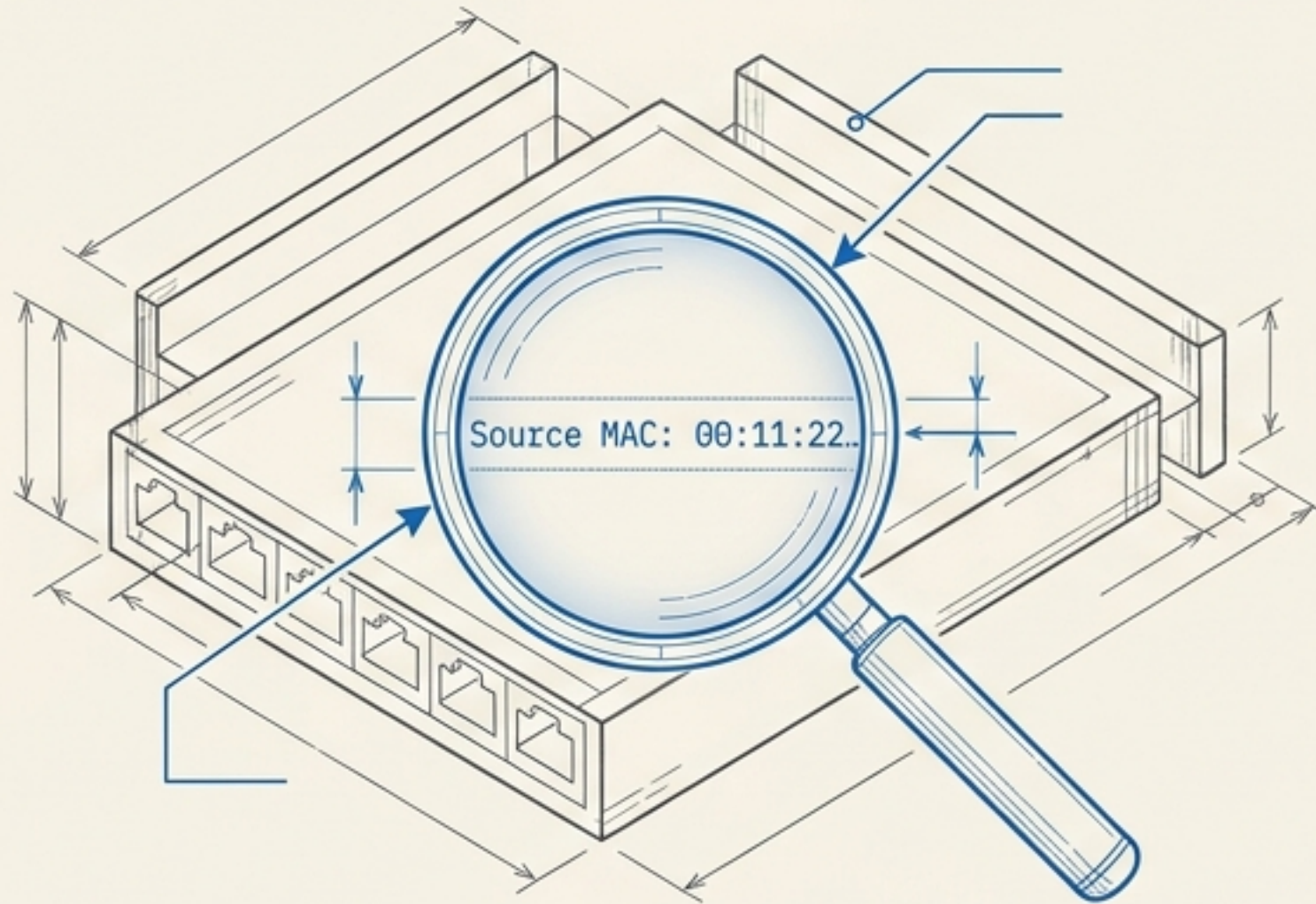
How Switches Learn, Forward, and Defend

A visual guide to CAM tables, flooding mechanisms, and port security.



The Golden Rule of Ethernet Switching

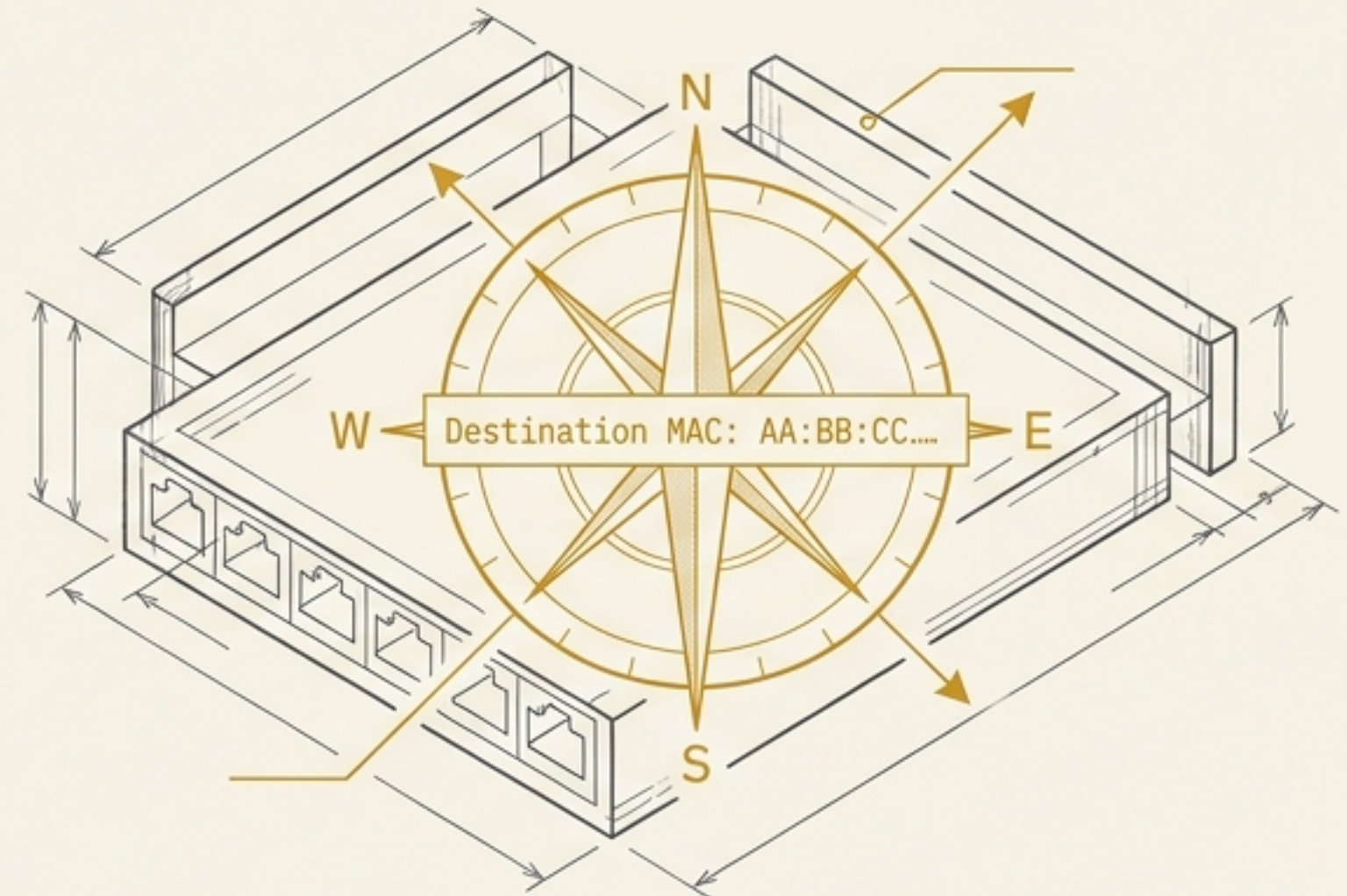
Learn (Source)



"Where did this come from?"
The switch uses the Source MAC
to map network locations.

Learning and forwarding are
completely independent
algorithmic operations.

Decide (Destination)



"Where does this need to go?"
The switch uses the Destination
MAC to determine the exit path.

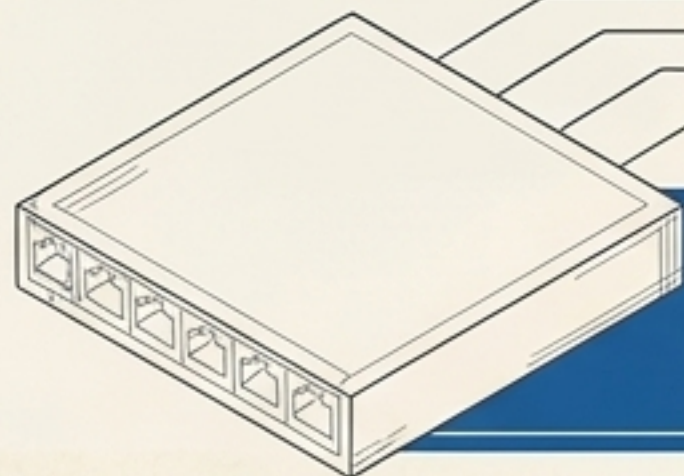
Anatomy of the CAM Table

VLAN Context	MAC Address	Port
10	00:11:22:33:44:55	Gi0/1
10	AA:BB:CC:DD:EE:FF	Gi0/2
20	10:20:30:40:50:60	Gi0/5

Decisions are bounded by Layer 2 domains.

The unique hardware identifier.

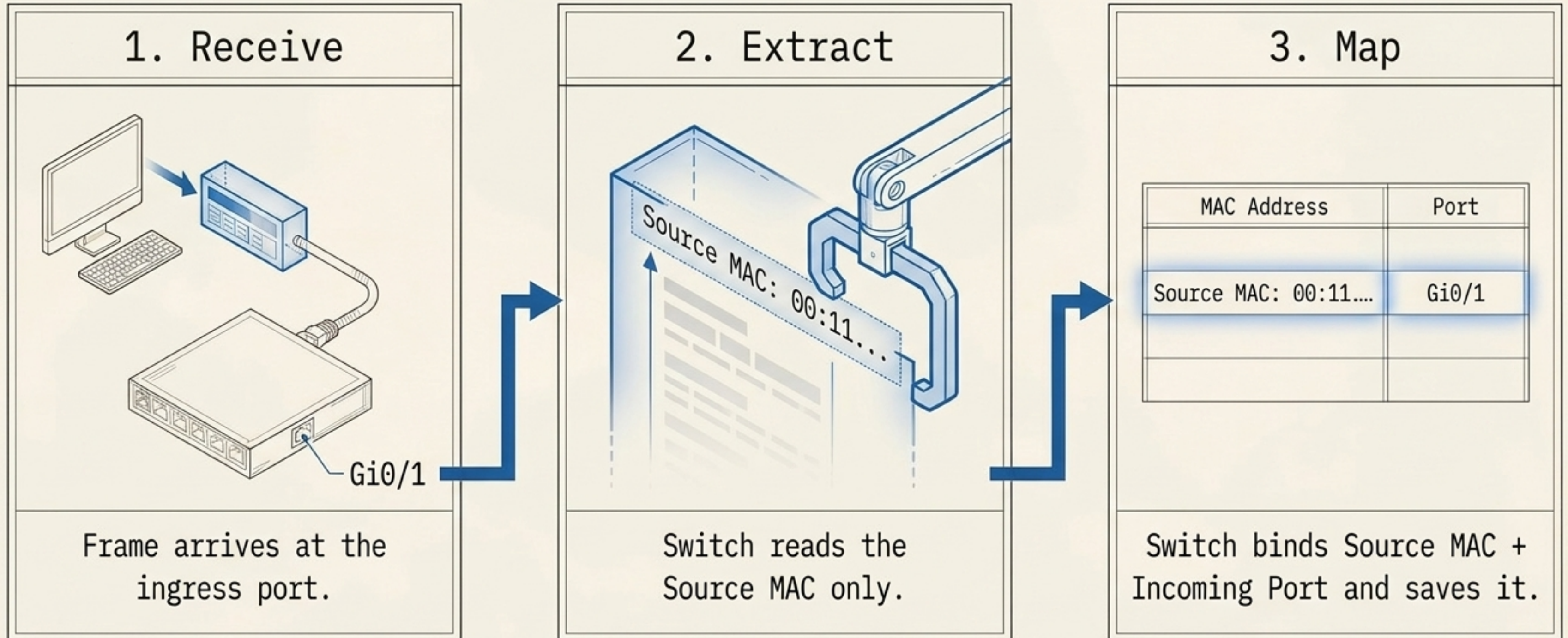
The physical interface for reachability.



The CAM table exists to answer one precise question:
"If I need to send a frame to this MAC address, which port should I use?"

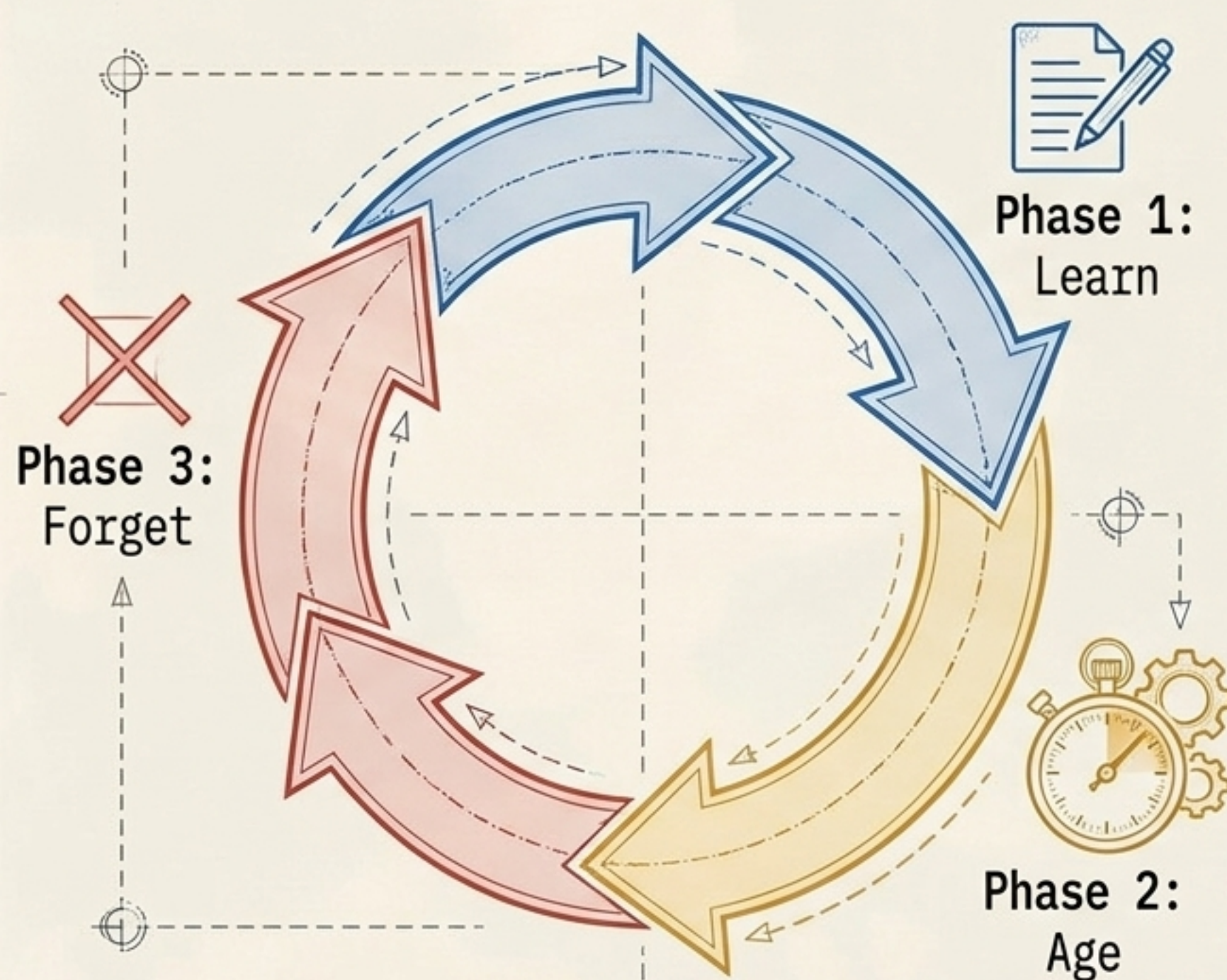
The Ingress Learning Mechanism

The switch ignores the destination MAC during the learning phase.



The Lifecycle of a Record

The CAM table is a dynamic map, not a permanent inventory.

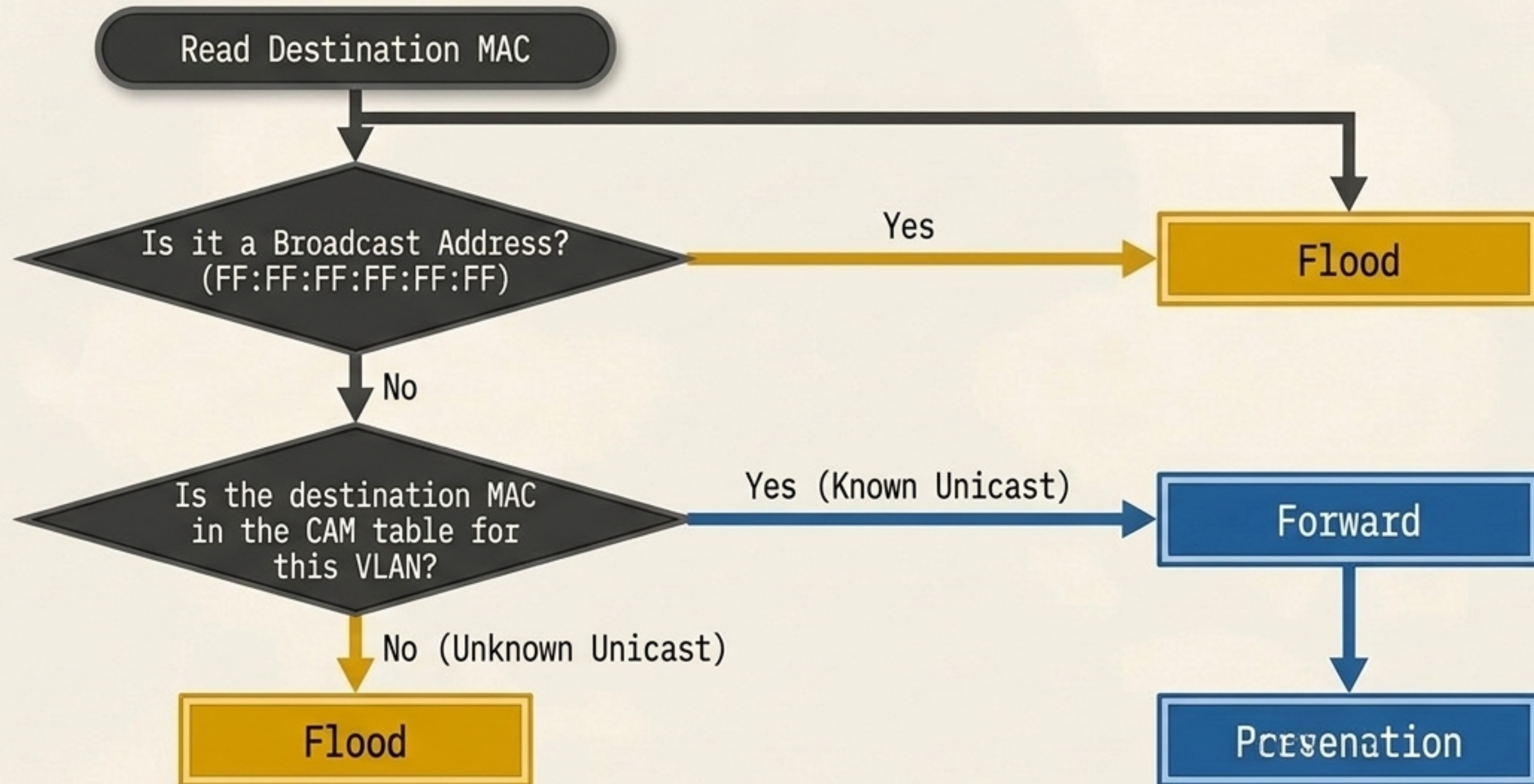


Aging Mechanism: Stale entries expire if a device is silent.

Adaptability: If a known MAC arrives on a new port, the switch instantly overwrites the old record.

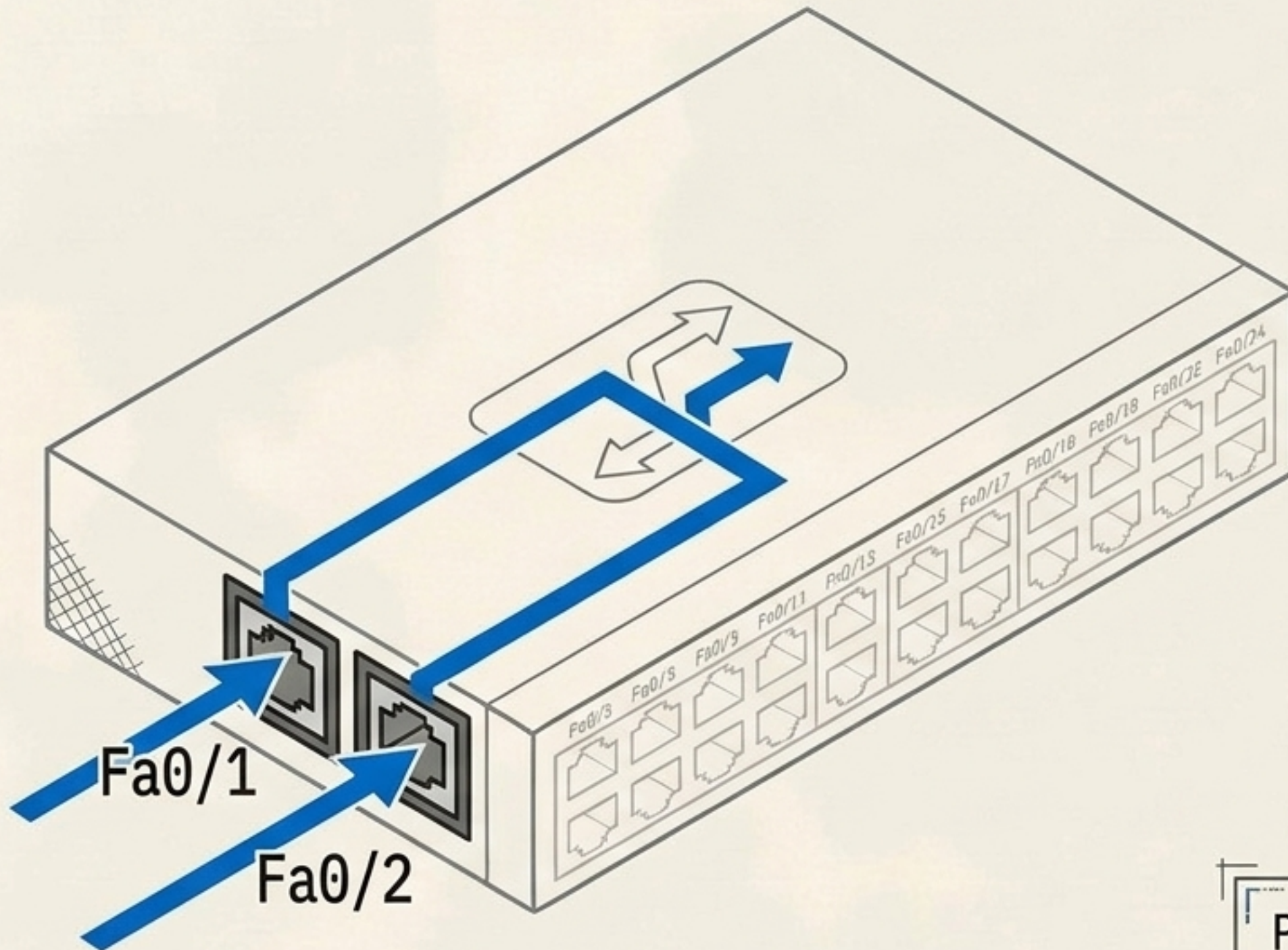
Purpose: Allows networks to adapt to moved devices, disconnected hardware, and topology changes.

The Master Decision Engine



Flooding is the switch's default failsafe when information is missing.

Known Unicast: Precision Forwarding



When the destination MAC is known, the switch uses "Filtering and Forwarding."

1. Look up Destination MAC.

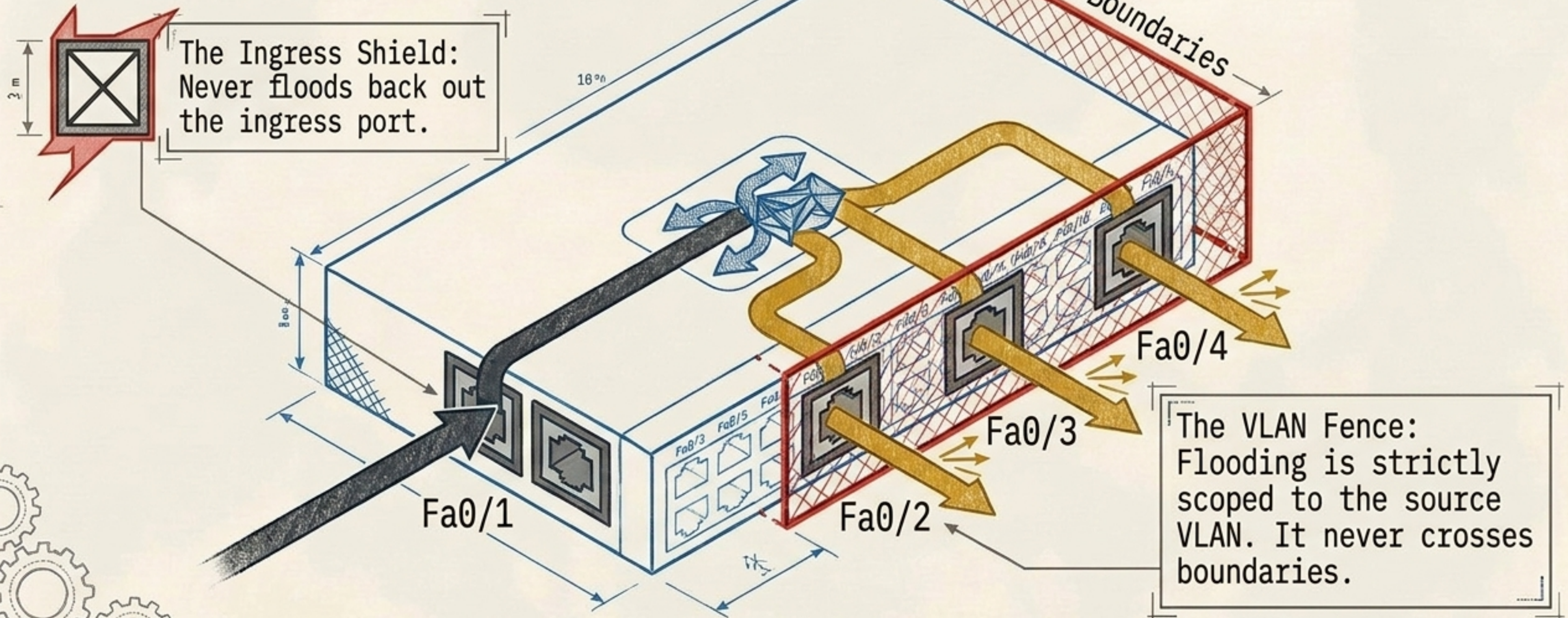
2. Identify specific Egress Port.

3. Filter the frame from all unrelated ports.

Preserves network bandwidth and provides baseline Layer 2 confidentiality.

The Flooding Radiator (Unknowns & Broadcasts)

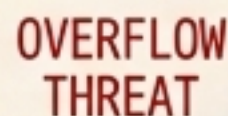
Unknown Unicast and Broadcasts are flooded to all eligible paths.



The Layer 2 Paradigm

Headers	Legacy Hub	Modern Switch
Layer of Operation	Layer 1 (Physical)	Layer 2 (Data Link)
Memory	None	Dynamic CAM Table
Traffic Handling	Repeats everything blindly	Forwards selectively (floods only when necessary)
Collision Domains	One shared domain	Independent domains per port

A switch may temporarily behave like a hub when flooding unknown unicasts, but it is fundamentally governed by intelligent, selective Layer 2 decision-making.



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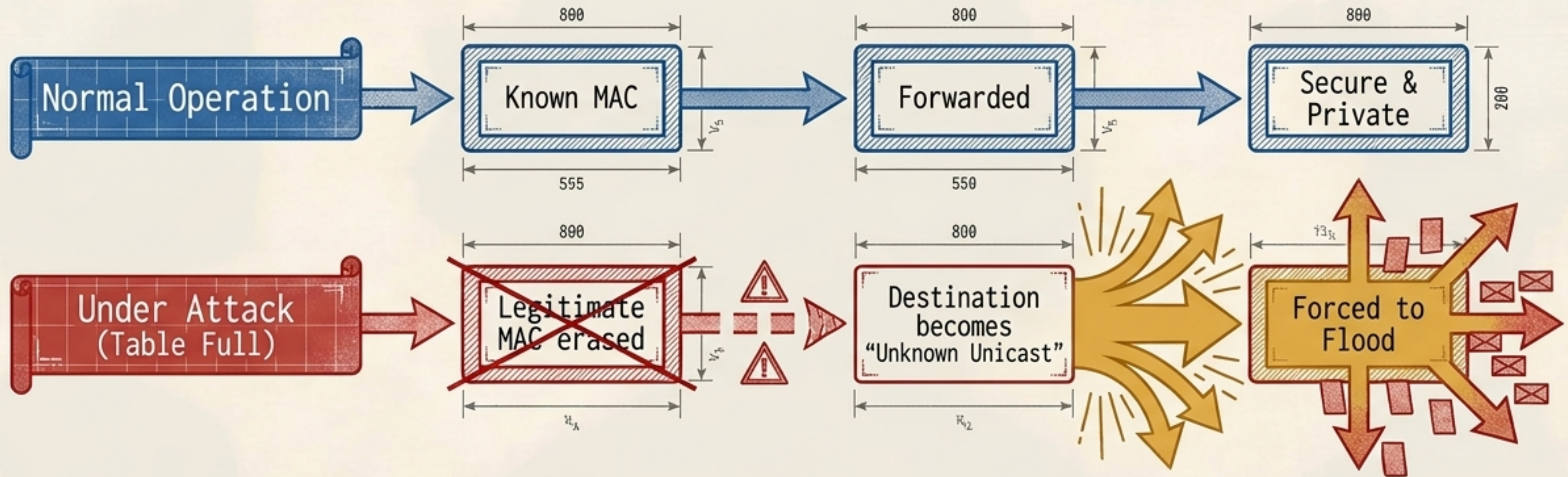
2.

Diagram illustrating a switch memory controller architecture. The system includes a SWITCH MEMORY CONTROLLER, an AUTOMATIC LEARNING block, and a memory array (CAM). Data is written to the CAM, as indicated by the large red arrow labeled "WRITING TO CAM". The diagram also shows input data (represented by red envelopes) being processed by the AUTOMATIC LEARNING block. Dimensions are marked: 1 cell for the controller and learning blocks, and 2 cells for the memory array. A horizontal dimension $R/2$ is also indicated.

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- LEGITIMATE ENTRIES

 Gemini Notebook

The Impact of Exhaustion



Denial of Service

Degrades switch performance and consumes link bandwidth.

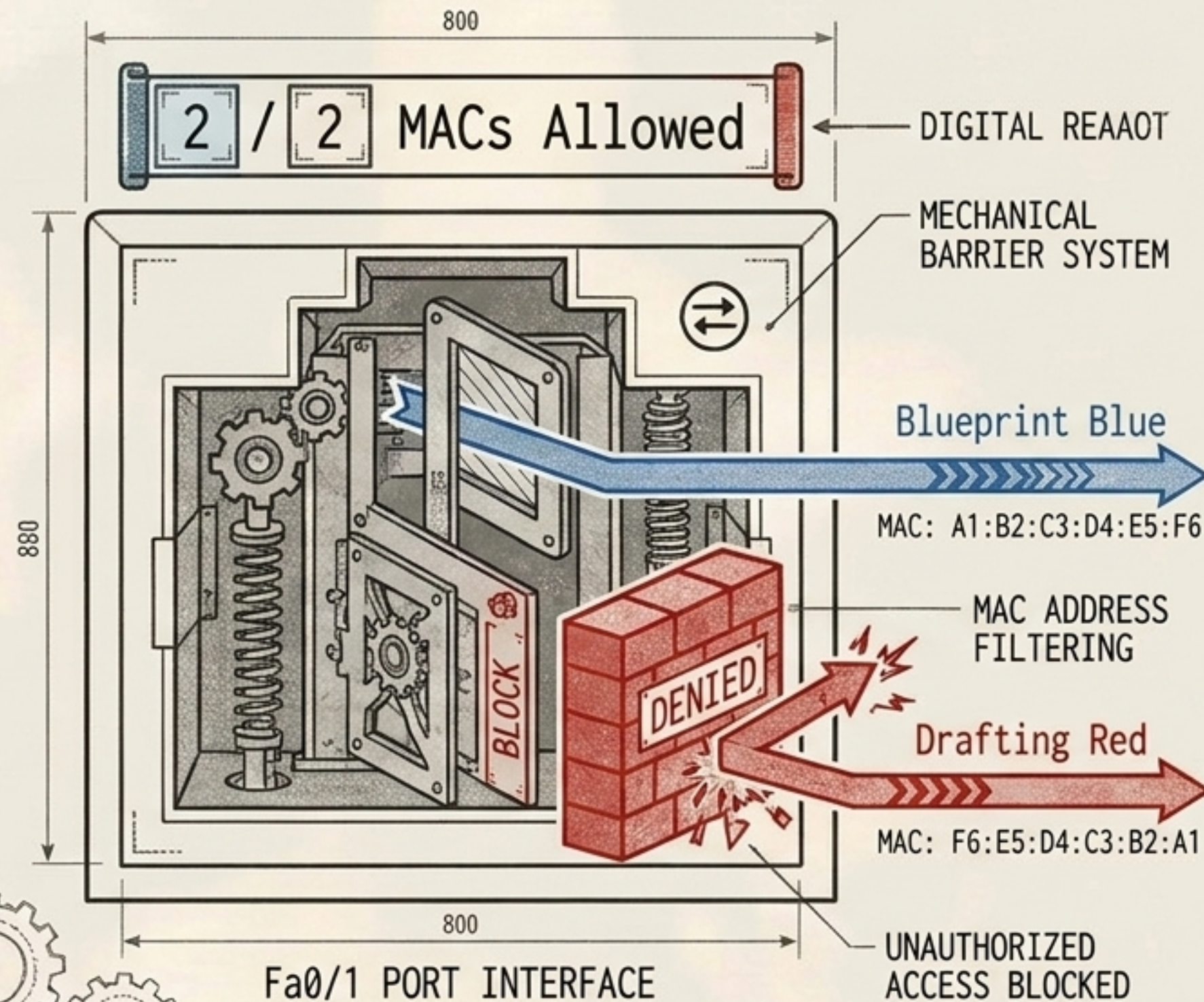
Increased Exposure

Forces private traffic to be flooded out to unrelated ports.

MAC flooding does not automatically guarantee full traffic capture.
It is a disruption technique that degrades the switch into flooding mode.

The First Line of Defense: Port Security

Port Security restricts the behavior of MAC addresses on an access port.



Maximum Limits:

Administrators define a hard limit (e.g., 1 or 2 unique MACs) per physical port.

Sticky MAC:

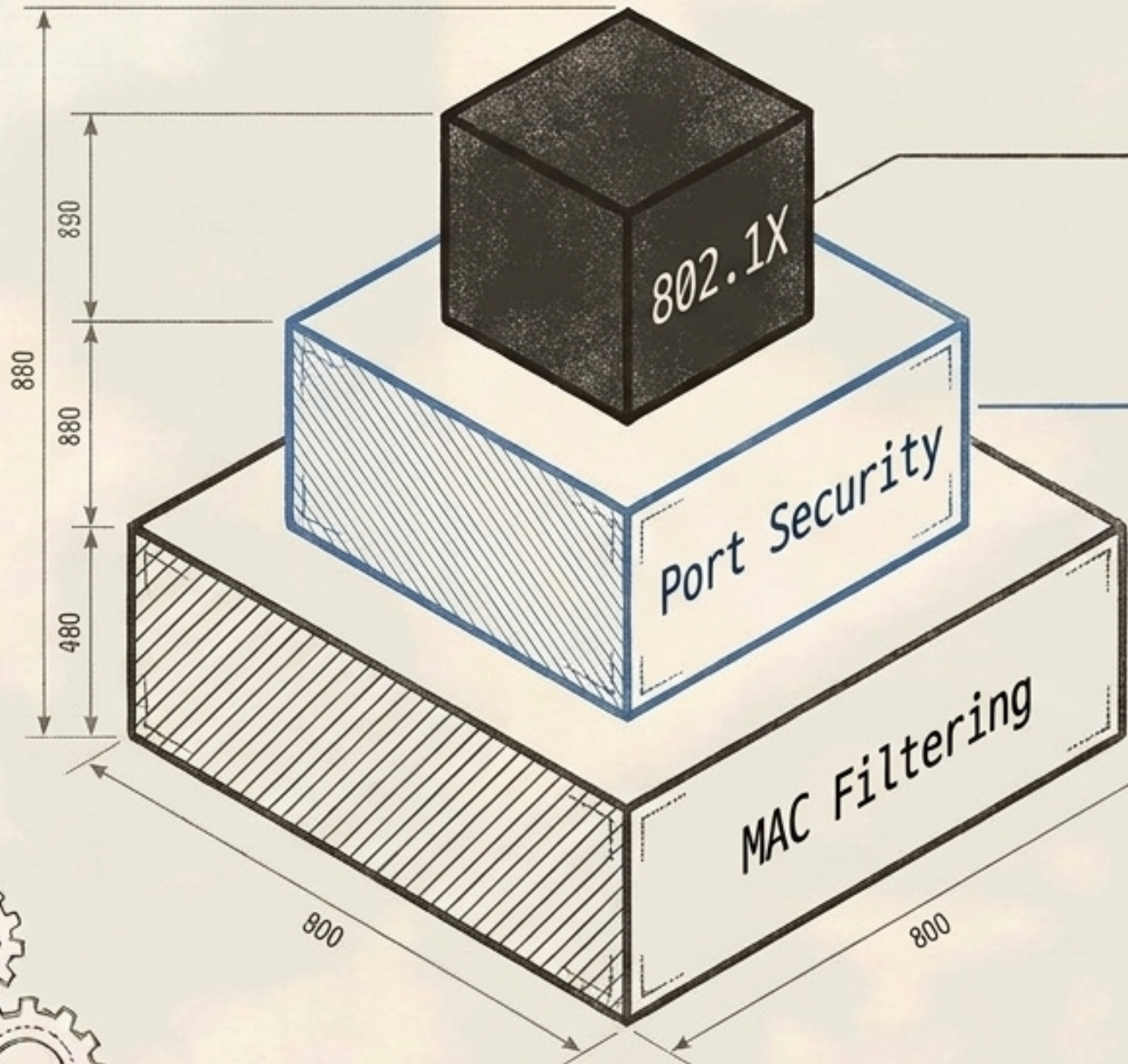
The switch can dynamically learn the first expected device and permanently lock it in as the authorized device for that port.

Port Security Violation Actions

What happens when the MAC limit is breached?



The Layer 2 Security Hierarchy



MAC Filtering

- Concept: Allow/Deny list of MACs.
- Flaw: Easily bypassed via MAC spoofing. An identifier, not proof of identity.

Port Security

- Concept: Limits maximum MACs per port.
- Strength: Highly effective against MAC Flooding. Controls behavior, not identity.

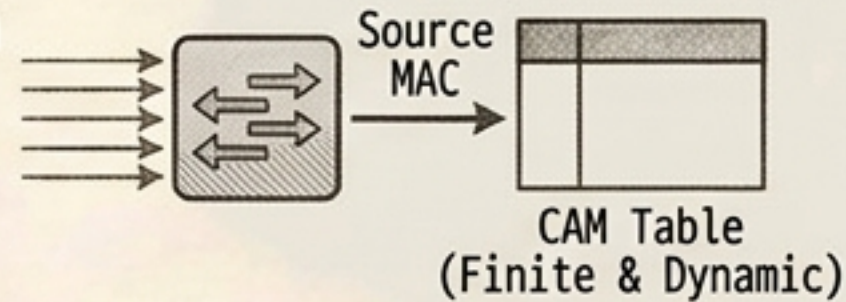
802.1X

- Concept: Port-based network access control.
- Strength: True cryptographic authentication. A forwarding restriction is not authentication.

The Layer 2 Blueprint (Cheat Sheet)

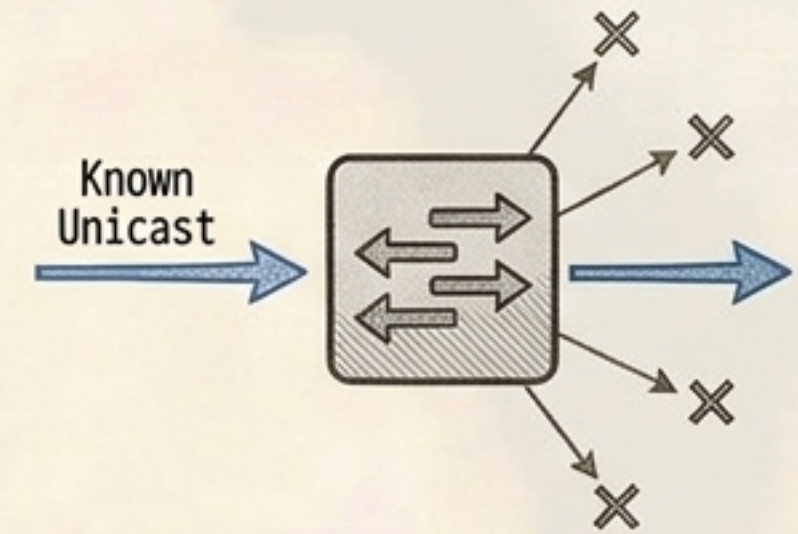
1. Learn

Switches learn from the Source MAC on the ingress port. CAM tables are finite and dynamic.



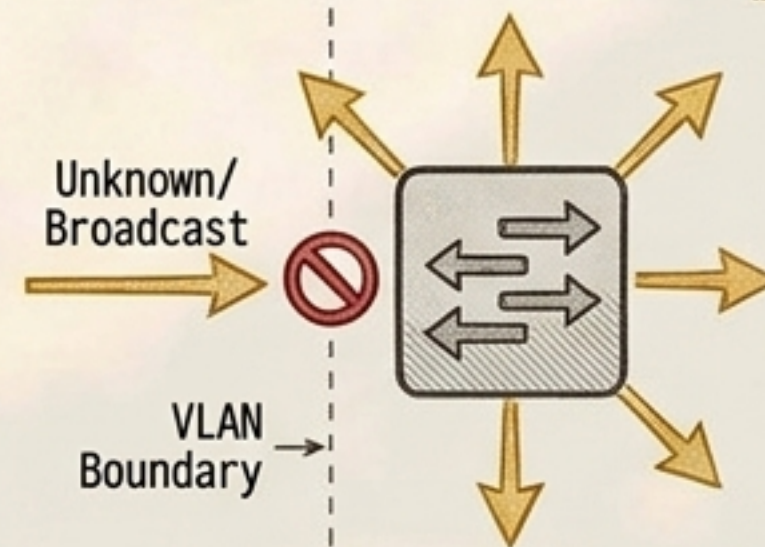
2. Forward

Known Unicast destinations are precisely filtered and forwarded.



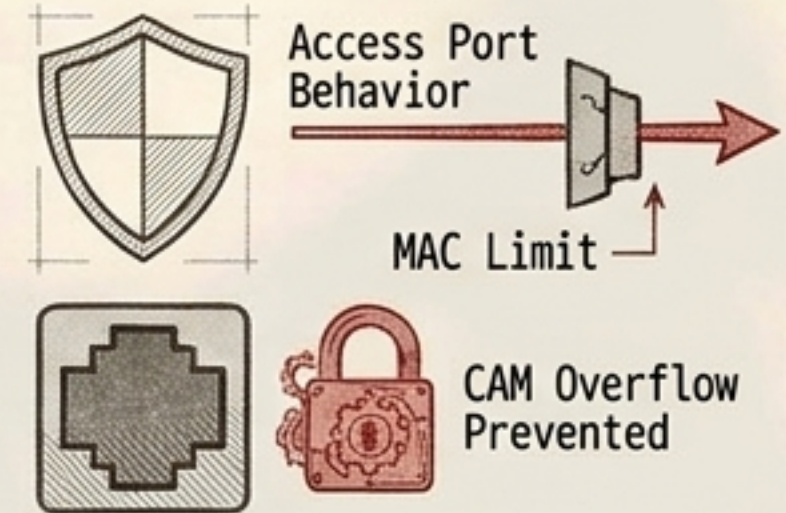
3. Flood

Unknowns & Broadcasts are flooded within their VLAN (never out the ingress port).



4. Defend

Port Security limits access port behavior to prevent CAM overflow attacks.



“Good network security controls the behavior that creates risk, rather than simply assuming an observed MAC address represents a trusted device.”